Prospective Restricted-Domain Validation of an Upward-Beam Spacecraft-Relative Potential Proxy in Lunar Prospector ER Data

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Abstract

An upward field-aligned electron feature provides the most stable voltage-scale observable recovered in our clean-room analysis of the Lunar Prospector Electron Reflectometer benchmark. A subsequent full-domain simulation study found calibrated held-out centroid inference at 150 and 350 eV but failed at 65 and 800 eV and exceeded its beam-absent false-detection gate. Because the candidate 100–500 eV operating window was proposed after those outcomes were known, it requires a new prospective experiment. We preregister a polarity-aware, sector-level validation in which continuum structure is independent of injected beam energy, calibration and confirmation use disjoint instrument supports, two normalizations are derived from a common latent observation, and response, morphology, temporal, semi-synthetic, null, and out-of-window challenges are explicit. The primary matrix contains 4800 moderate-to-strong beams over twelve energy-amplitude cells, accompanied by 600 weak beams, 1200 semi-synthetic injections, 3000 nulls, and 1000 out-of-window classification controls. Retention is co-primary with conditional interval coverage, and every cell must pass. A full pass would validate selective measurement of a pre-response beam centroid over the tested envelope and open a hand-audited one-day real-data study. It would not validate absolute lunar surface potential, spacecraft potential, loss-cone corroboration, or a global map. No simulation result had been generated when this design was written.

1 Scientific Background

For a surface-emitted electron transported electrostatically to the spacecraft, energy conservation gives, in electron-volt/volt numerical units,

$$E_{\mathrm{beam}} \simeq E_{\mathrm{emit}} - U_{\mathrm{rel}} + \delta_{\mathrm{transport}}, \qquad U_{\mathrm{rel}} = U_{\mathrm{M}} - U_{\mathrm{sc}}. \tag{1}$$

The commonly used proxy $U_{\mathrm{rel}} \simeq -E_{\mathrm{beam}}$ sets the initial secondary energy and transport discrepancy to zero. This is often a useful leading-order scale, but it is not an identity. Absolute lunar potential additionally requires an independent U_{sc} estimate:

$$U_{\mathrm{M}} = U_{\mathrm{rel}} + U_{\mathrm{sc}}. \tag{2}$$

Experiment 004 found a robust response-aware beam scale near 145–160 eV in the historical benchmark but did not find stable independent voltage information in the loss-cone boundary. Experiment 005 therefore tested beam-centroid inference directly. Its smooth-continuum estimator achieved calibrated held-out results in both 150 and 350 eV cells, but weak low-energy and both high-energy conditions failed; beam-absent acceptance was 6% against a frozen 5% point-estimate gate. The complete 20–2000 eV hypothesis failed.

The observed middle-energy success motivates a restricted hypothesis. It does not confirm one. The present experiment is designed so that the new window can be rejected without another retrospective rescue.

2 Lessons From The Parent Experiment

Several limitations became visible only when the complete Experiment 005 record was reconsidered.

First, the synthetic loss-cone boundary was generated using $U_{\mathrm{rel}} = -E_{\mathrm{beam}}$. The nuisance continuum therefore contained indirect information about the injected answer even though the fitted continuum itself had no voltage equation. The new primary generator samples continuum parameters independently of beam energy.

Second, both normalization branches received the same already-normalized synthetic observation. Their agreement was automatic. The new experiment starts from one latent native-sector flux realization and derives incident-mean and conjugate-sector normalizations independently.

Third, pitch was computed without establishing which magnetic-field direction connected to the surface. Here surface intersection and polarity are required before defining 180 degrees as surface-leaving. Disconnected geometry is excluded rather than assigned a directional interpretation.

Fourth, four support spectra were shared by calibration and validation, coverage was conditional on selection without a strong retention requirement, and the null point estimate was underpowered as an upper-bound statement. The new design uses disjoint outcome-blind support sets, makes retention co-primary, and uses exact binomial confidence limits.

3 Statistical Estimand

Let $S = 1$ denote passage of every frozen geometry, numerical, normalization, stability, and interval-resolution rule. The primary estimands are

$$r_j = \Pr(S = 1 \mid j), \tag{3}$$

$$b_j = \mathrm{median}\left[\frac{\widehat{E}_b - E_b}{0.48E_b}\middle| S = 1, j\right], \tag{4}$$

$$c_{q,j} = \Pr(E_b \in I_q \mid S = 1, j), \qquad q \in \{0.68, 0.95\}, \tag{5}$$

for each prespecified energy-amplitude cell j. Reporting coverage only conditional on $S = 1$ is legitimate only when r_j is reported and gated. Every simulation, including a crash or unresolved fit, remains in the retention denominator.

For a beam-absent family k, define

$$p_{0,k} = \Pr\left(S = 1,\ 100 \le \widehat{E}_b \le 500 \mid \text{no beam}, k\right). \tag{6}$$

The gate is not $\widehat{p}_{0,k} \le 0.05$. It requires the Bonferroni-adjusted one-sided familywise-95% Clopper–Pearson upper bound to be at most 0.05. With three families of 1000 controls, no family may contain more than 35 false in-window acceptances.

4 Estimator And Reporting Domain

The observed support and optimizer remain broad, 20–2000 eV. The reporting window, 100–500 eV, is applied after fitting. This separation prevents a null or out-of-window feature from being forced

against a newly narrowed optimizer bound. Confidence intervals are not clipped at 100 or 500 eV, and base QC is computed without conditioning on whether the fitted point falls inside the window.

The model is

$$R(E, \alpha) = C(E, \alpha) + AG_E(E; E_b)G_\alpha(\alpha; 180^\circ, \sigma_\alpha). \tag{7}$$

The pitch-logistic continuum C has cone level, contrast, boundary, and width varying quadratically in normalized log energy. It includes no adiabatic boundary equation. The nominal energy response has FWHM$/E = 0.48$ and is integrated over channel support. The fitter uses the frozen unweighted log-residual objective, deterministic multistart, and quality checks inherited in concept from Experiment 005.

5 Support And Information Barriers

Twenty-four supports are selected from the 398-sweep parent interval using only time, structural validity, connection, pitch coverage, native energy alignment, valid-sector counts, and magnetic variation. Known Experiment 005 supports and the Halekas neighborhood cannot enter confirmation. Twelve supports are assigned to calibration and twelve disjoint supports to confirmation.

The experiment has four barriers. First, support selection and geometry are committed. Second, implementation verification and fresh calibration are run. Third, code, corrections, seeds, checksums, and decision logic are frozen. Fourth, confirmation runs once without incremental outcome inspection. Any method change after that point requires unused draws and a new experiment.

6 Calibration And Confirmation

Calibration uses 3000 beams at 100, 140, 220, 340, and 500 eV. One deterministic half estimates median bias and the other estimates rank-based 68% and 95% interval multipliers. No Experiment 005 row or correction value is reused.

Confirmation crosses six energy strata—the exact endpoints and four log-uniform interior bands—with amplitudes 4–8 and 8–16. Four hundred cases in each cell give 4800 primary beams. Pitch width, noise, support, continuum, response, and channel phase are balanced independently within every cell. Six hundred amplitude-1–4 beams characterize, but cannot expand, the primary claim.

The primary generator includes independent smooth, double-transition, asymmetric, and correlated energy-pitch backgrounds. Challenges include heteroskedastic and correlated noise, channel integration, response-width and energy-scale stress, log-energy and asymmetric response, beam-axis offsets, double peaks, missing channels, and temporal changes during the sequential energy sweep. A further 1200 injections use held-out real residual templates.

Three thousand beam-absent controls cover matched, misspecified, and adversarial families. The adversarial family includes energy-only and pitch-only enhancements, single-row spikes, incident-side features, and temporal ramps or pulses. One thousand true out-of-window beams test leakage into the reporting window; the 500 cases on each side may produce no more than 15 final-valid in-window reports under their adjusted exact-binomial gate.

7 Decision Rule

Every one of twelve primary cells must retain at least 200 of 400 attempts. Among retained cases, the bootstrap upper confidence bound for median absolute error must not exceed 0.10 response

FWHM, the corresponding bound for the 90th percentile must not exceed 0.25 FWHM, and the signed-median interval must lie within ±0.05 FWHM. Empirical 68% and 95% coverage must lie in 0.60–0.76 and 0.90–0.99, respectively, with each nominal level contained in its exact 95% binomial interval. Every retained 95% interval must be no wider than one response FWHM. Pooled background-family gates, semi-synthetic gates, null gates, and out-of-window leakage gates must also pass.

These requirements form an intersection claim: the complete window passes only when all required components pass. Failure at an edge rejects the full window. Failure of null control may leave centroid recovery interesting for an independently established beam, but it does not validate automated detection. Failure of the semi-synthetic layer limits any success to the simulation system.

8 Expected Outcomes

Three outcomes are scientifically plausible. First, the middle-energy success may generalize across new supports and energy locations, including the proposed edges. This would justify a one-day, hand-audited observable study. Second, the central regime may remain valid while one or both edges fail. This rejects the 100–500 eV claim and supplies an exploratory hypothesis for a later experiment; it is not a partial pass. Third, response, polarity, normalization, temporal, or semi-synthetic challenges may reveal that the earlier success was too simulator-specific. That negative result would prevent a misleading real survey and identify the missing physics or observation model.

9 Interpretation Limits

A full pass validates a selective measurement procedure for E_{beam} over its tested amplitude, morphology, support, and discrepancy envelope. It does not show that every accepted real feature is a lunar secondary, explain beam absence, measure real-data bias without truth, validate U_{sc}, recover U_{M}, provide independent loss-cone corroboration, establish mission-wide representativeness, or reverse the global reconstruction's identifiability limit.

The real-data stage, if opened, is therefore a residual and prevalence audit. Its purpose is to learn whether the validated observable occurs in actual LP ER sweeps under the same quality rules, not to jump directly from a successful simulation to a lunar potential map.

10 Current Status

This document records the design before support selection, implementation, calibration, or confirmation. Numerical response-stress values, deterministic seed hashes, the support manifest, and the semi-synthetic template manifest must be frozen in a later pre-calibration commit. No Experiment 006 scientific result exists at this stage.